



UPPSALA
UNIVERSITET

ANALYSIS SUBSECTION

Is the Best, the Best? Analysing Best-Paper Awardees' Career Trajectories

Student:

Shaheryar Shaheryar
shaheryar.4822@student.uu.se

Supervisors:

Georgios Panayiotou
georgios.panayiotou@it.uu.se

Subject Reviewer:

Arjun Menon
arjun.menon@it.uu.se

Department Of Information Technology
March 12, 2026

Contents

1	Introduction	3
2	Related Work	4
3	Methods	4
4	Analysis	5
4.1	Best Paper Awards Accelerate Junior Careers, But Leave Senior Co-Authors Unchanged	5
5	Discussion	7
6	Conclusion	7

1 Introduction

At pretty much every computer science conference, a handful of papers get singled out as "best." Committees deliberate, some teams walk up on stage, and everyone adds a new line to their CV. But what actually happens after that? Does winning change anything for the junior researchers involved, or does it mostly just confirm what was already going to happen anyway.

Recognition matters a lot in how science works. Tenure decisions, grant competitions, hiring, all of these lean heavily on signals like publications, citations, and prizes. And informally too, an award changes how people read your work, how likely collaborators are to reach out, how your application looks in a pile. But the relationship between getting recognised and what comes next isn't obvious. Studies of big prizes like the Nobel or Fields Medal actually show that winning can sometimes be followed by lower citation impact per paper, like being celebrated changes what people choose to work on [1, 2]. Other work finds the opposite that prize winners publish more and stay active longer than matched peers who didn't win. So awards might be both a marker of past excellence and something that actually reshapes what comes next.

This raises some uncomfortable questions about fairness. If early recognition accelerates some careers while others with similar potential never get that push, then awards might be amplifying inequalities more than just recording achievement. Or maybe the effect is small and awards mostly just find people who were already on a good path. Either way it matters not just for the researchers themselves but for universities, funders, and conference organisers who use these awards in evaluation all the time.

Most of the research on this stuff looks at rare, high-profile prizes for already established scientists. Computer science is different. Conferences¹ are where most of the field publishes, and best paper awards happen far more often than global prizes. They often go to teams with PhD students or recent graduates, right when careers are still being shaped. For a junior researcher, a best paper award might be the first real signal that their work is landing. But we don't actually know much about whether those awards are followed by meaningfully different career paths. Do junior awardees go on to publish more, get cited more, do more disruptive work or do their careers look about the same as similar people who just didn't happen to win.

This thesis tries to answer that. It looks at the career trajectories of junior authors who won best paper awards at CS conferences and compares them with matched non-winning peers from the same venues and years. "Junior" here means within the first few years of a researcher's indexed publication record the period where early opportunities might have long-lasting effects. By "career trajectory" we mean things you can actually measure papers per year, citations, how disruptive the contributions are, and how collaboration

¹<https://www.acm.org/conferences>

networks change over time. These are imperfect proxies for quality or creativity, but they're what actually shapes how careers get evaluated in practice.

The study builds on a few lines of existing research. Work on early-career recognition suggests that getting prizes earlier tends to amplify future achievement more papers, more citations, longer active careers, even after accounting for initial performance [3]. Research on what predicts long-term impact shows that a small set of early-career features (like citations, institutional prestige, senior collaborators) explain a lot of variation across fields [4]. And work on team structure shows that small teams are more likely to produce disruptive work while larger teams tend to consolidate existing lines [5]. Any effect of conference awards on junior researchers is probably tangled up with all of these things.

Against that background, the thesis asks three main questions. First, how do the publication and citation trajectories of junior awardees differ from similar non-awardees from the same conferences and years. Second, do awardees and non-awardees diverge in how disruptive their subsequent work is, using citation-based indices that capture whether papers redirect or consolidate the literature. Third, are any differences consistent across venues and career stages, or concentrated in particular contexts. There's also an optional fourth question about co-authorship networks and whether the structure of collaborations evolves differently for awardees over time.

The data comes from OpenAlex² combined with curated best paper award lists. By comparing awardees with carefully matched non-awardees the thesis aims to give an empirical picture of what conference-level recognition actually does to junior researchers' careers in CS and what that might mean for how we design and interpret these recognition systems.

2 Related Work

3 Methods

This study takes a quantitative, observational approach to compare the careers of junior best-paper awardees with those of similar non-awardees. The context is computer science, where conferences are the main publication venues and career progression is mostly evaluated through bibliometric indicators rather than anything you could observe directly in a workplace. So data comes from large-scale scholarly databases and curated award lists.

The main data source is OpenAlex. For a set of computer science conferences and years, we pull metadata on all papers and their authors including publication year, venue, citation counts, reference lists, and authorship. These records are then linked to curated best paper award lists from ACM and community-maintained collections, which give us conference,

²<https://docs.openalex.org>

year, title, authors, and award type for each recognised paper. Matching those award entries to OpenAlex records gives us a cohort of awarded papers and the junior authors on them.

The main units of analysis are authors who receive at least one best paper award within five years of their first indexed publication (junior awardees) and a matched set of non-awarded junior authors from the same conferences and years. Matching is done on venue, award year, and early-career stage, using nearest-neighbour matching on pre-award productivity and citations. For each author we build a longitudinal record of outputs and collaborations before and after the award year (or the corresponding year for controls).

The key variables are annual publication count as a measure of productivity, annual citation count and field-normalised citation rates for impact, team size and co-author diversity for collaboration, and a disruption score based on the CD index which captures how often later papers cite a focal work without also citing its references, a signal that the work redirected rather than extended the literature [6]. All of these are computed for several years before and after the award event so trajectories can be compared between groups.

For statistical analysis we combine descriptive summaries with inferential tests. Mann-Whitney U tests are used to compare median productivity and citation measures between groups, and regression models controlling for pre-award performance check whether award status is associated with different post-award trajectories. Where the data allows, we also run stratified analyses by venue and career stage to look for heterogeneity. Everything is implemented in Python with open-source libraries, and code and derived datasets are documented for reproducibility.

On the ethics side, only publicly available bibliometric metadata and award lists are used. No sensitive personal data is collected. Limitations around author disambiguation and database coverage are documented throughout, and results are interpreted as patterns in visibility and impact rather than judgments about individual researchers.

4 Analysis

4.1 Best Paper Awards Accelerate Junior Careers, But Leave Senior Co-Authors Unchanged

The data shows a clear asymmetry. Junior researchers on award-winning papers show meaningful career acceleration after the award, while their senior co-authors on the exact same paper do not. This matters because if the effect were just about paper quality or venue prestige, both groups should move together. The fact that they don't points to something specific about career stage. To get at this, we use a within-paper design,

meaning we compare junior and senior co-authors of the *same* award-winning paper. This is actually a pretty strong setup, because it controls for paper quality, venue, and year all at once. If juniors and seniors on the same paper end up on different trajectories afterwards, it’s hard to explain that away.

To measure career change we use something called *lift*, basically the ratio of someone’s average annual citations (or publications) in the three years after the award compared to the three years before it. A lift of 1.0 means nothing changed. Above 1.0 means things got better. For every award paper, we identify the junior authors (career age ≤ 5 years at time of award) and the senior co-authors (career age > 5 years), and compute their individual lift scores.

Figure 1 shows the distribution of citation lift and publication lift for 603 junior awardees and 1,189 senior co-authors, across 31 computer science conferences from 2000–2018. The pattern is pretty clear. Junior researchers have a median citation lift of around **1.5**, so their annual citations went up by roughly 50% after the award. Senior co-authors on the exact same papers have a median lift of around **1.0**, basically no change at all. The same thing shows up for publications, juniors are around 1.7, seniors are near baseline. Mann–Whitney U tests confirm both differences are statistically significant (citations: $p < 0.001$, publications: $p < 0.001$).

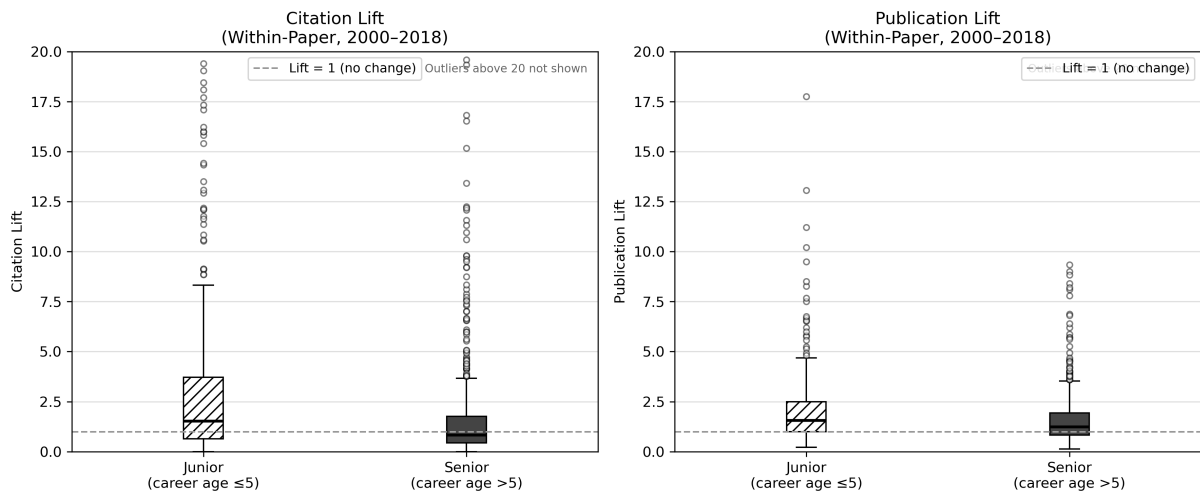


Figure 1: Citation lift (left) and publication lift (right) for junior (career age ≤ 5) and senior (career age > 5) co-authors on the same award-winning papers. The dashed line at lift = 1.0 marks no change from pre- to post-award. Outliers above 20 are omitted for readability.

The wide spread among juniors is also worth noting. While the median is moderate, there’s a long tail of junior researchers with very large lifts, citation lifts of 5, 10 or even higher. So the award effect isn’t the same for everyone. For some junior researchers it looks genuinely career-changing, for others the boost is pretty small. Seniors on the other hand are bunched tightly around 1.0, which makes sense their careers are already

established and one award probably isn't going to move the needle much.

This fits with what Zhu et al. [3] find, that earlier recognition amplifies future achievement more than recognition received later. What we add here is a cleaner test of that idea, using the award paper itself as the control. The award does seem to matter it just matters a lot more depending on where you are in your career when it happens.

5 Discussion

6 Conclusion

References

- [1] Andrew Nepomuceno, Hilary Bayer, and John P. A. Ioannidis. “Impact of major awards on the subsequent work of their recipients”. en. In: *Royal Society Open Science* 10.8 (Aug. 2023), p. 230549. ISSN: 2054-5703. DOI: 10.1098/rsos.230549.
- [2] George Borjas and Kirk Doran. *Prizes and Productivity: How Winning the Fields Medal Affects Scientific Output*. en. Tech. rep. w19445. Cambridge, MA: National Bureau of Economic Research, Sept. 2013, w19445. DOI: 10.3386/w19445.
- [3] Wanying Zhu et al. “Earlier recognition of scientific excellence enhances future achievements and promotes persistence”. en. In: *Journal of Informetrics* 17.2 (May 2023), p. 101408. ISSN: 17511577. DOI: 10.1016/j.joi.2023.101408.
- [4] Alexander Krauss, Lluís Danús, and Marta Sales-Pardo. “Early-career factors largely determine the future impact of prominent researchers: evidence across eight scientific fields”. en. In: *Scientific Reports* 13.1 (Nov. 2023), p. 18794. ISSN: 2045-2322. DOI: 10.1038/s41598-023-46050-x.
- [5] Lingfei Wu, Dashun Wang, and James A. Evans. “Large teams develop and small teams disrupt science and technology”. en. In: *Nature* 566.7744 (Feb. 2019), pp. 378–382. ISSN: 0028-0836, 1476-4687. DOI: 10.1038/s41586-019-0941-9.
- [6] Joerg Sixt and Michele Pasin. “Dimensions: Calculating disruption indices at scale”. en. In: *Quantitative Science Studies* 5.4 (Nov. 2024), pp. 975–990. ISSN: 2641-3337. DOI: 10.1162/qss_a_00328.